5.5.2 Electromagnetic radiation from stars

	Learning outcomes	Additional guidance
	Learners should be able to demonstrate and apply their knowledge and understanding of:	
(a)	energy levels of electrons in isolated gas atoms	
(b)	the idea that energy levels have negative values	
(c)	emission spectral lines from hot gases in terms of emission of photons and transition of electrons between discrete energy levels	HSW2, 8
(d)	the equations $hf = \Delta E$ and $\frac{hc}{\lambda} = \Delta E$	Learners will also require knowledge of section 4.5
(e)	different atoms have different spectral lines which can be used to identify elements within stars	

(f)	continuous spectrum, emission line spectrum and absorption line spectrum	
(g)	transmission diffraction grating used to determine the wavelength of light	The structure and use of an optical spectrometer are not required; **PAG5**
(h)	the condition for maxima $d\sin\theta = n\lambda$, where d is the grating spacing	Proof of this equation is not required.
(i)	use of Wien's displacement law $\lambda_{max} \propto \frac{1}{T}$ to estimate the peak surface temperature (of a star)	*M0.4* HSW5
(j)	luminosity L of a star; Stefan's law $L = 4\pi r^2 \sigma T^4$ where σ is the Stefan constant	Learners will also require knowledge of 4.4.1
(k)	use of Wien's displacement law and Stefan's law to estimate the radius of a star.	*M0.4* HSW5